Two negative findings about $B^{(\lambda)}_{\ell+1}V_\lambda$: non-stable dimension drop and JM-non-semisimplicity

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Abstract

We retain the setup of the May-13 papers: $\lambda \vdash n$ with ℓ rows, $R'_j := (T_{j-1}+1)$ acting on V_λ in the Hoefsmit seminormal basis, and

$$B \;:=\; B^{(\lambda)}_{\ell+1}V_\lambda \;:=\; \operatorname{im}\bigl(R'_{\ell+1}R'_{\ell+2}\cdots R'_n\bigr)\,.$$

The stable-regime theorem says $\dim B = f^{\lambda^{(\geq 2)}}$, where $\lambda^{(\geq 2)}$ is λ with its first column deleted. This note records two negative findings from a wider computational scan.

Finding 1. The SYT count $f^{\lambda^{(\geq 2)}}$ is *not* a universal upper bound on $\dim B$. At $\lambda = (3,3,2,2)$, $n = 10$, the computation gives $\dim B = 6 < 9 = f^{(2,2,1,1)}$. Conversely the formula *overshoots* in other non-stable shapes (e.g. a factor of 2 for shapes with paired equal non-trivial rows: $(4,2,2,1)$ and $(4,4,1,1)$).

Finding 2. B is not Jucys–Murphy semisimple. For $\lambda = (3,2,1,1,1)$ ($n = 8$, $\dim B = 2$) and $\lambda = (3,2,1,1)$ ($n = 7$, $\dim B = 2$), the Hoefsmit support of B is broad (26 of 64 SYTs and 23 of 35 SYTs respectively), and only a small number of J_k and T_j preserve B. Hence B is not a direct sum of JM weight spaces, and no candidate iso $B \cong V_{\lambda^{(\geq 2)}}$ can work by exhibiting a subset-of-SYTs bijection in the seminormal basis.

1 Setup

We use the conventions of the two May-13 papers on multi-corner dimension and the failed decoration iso. Briefly: $H_q(S_n)$ is the Iwahori–Hecke algebra; V_λ is the Specht module in the Hoefsmit seminormal basis $\{v_T : T \in \mathrm{SYT}(\lambda)\}$; the Jucys–Murphy elements are

$$J_k \;=\; \sum_{i<k}(i,k)_q,$$

the standard q-deformed sum of transpositions, which act diagonally on the Hoefsmit basis with eigenvalues given by q-contents along T. For λ with ℓ rows we set

$$B \;:=\; B^{(\lambda)}_{\ell+1}V_\lambda \;=\; \operatorname{im}\bigl(R'_{\ell+1}R'_{\ell+2}\cdots R'_n\bigr)\,, \qquad R'_j := T_{j-1}+1.$$

Throughout, $\lambda^{(\geq 2)}$ denotes the partition obtained by deleting column 1 of λ, i.e. $\lambda^{(\geq 2)} = (\lambda_1 - 1, \lambda_2 - 1, \ldots)$ with zero parts dropped. All computations are mod $P = 100\,003$ at $q = 11$ (a non-special value) via the seminormal-matrix infrastructure of `compute_dim_B.py`; mod-P rank equals rank over $\mathbb{Q}(q)$ for non-special q.

2 Finding 1: the non-stable regime

The May-13 closed-form theorem gives $\dim B = f^{\lambda^{(\geq 2)}}$ under a stability hypothesis (loosely: sufficiently many trailing 1-rows). It is tempting to conjecture that, even outside the stable regime, $\dim B \leq f^{\lambda^{(\geq 2)}}$ holds universally — the SYT count being a natural "upper envelope" from the recursive picture. The present finding refutes that conjecture in both directions.

2.1 The scan

We computed $\dim B$ for all partitions $\lambda \vdash n$ with $n \leq 11$, $\ell(\lambda) \geq 3$, and $\lambda_1 \geq 2$ (i.e. at least two columns and at least three rows — the regime where B is non-trivial and the column-1 deletion is non-degenerate). The table below gives a representative slice; the full output is in `run_nonstable.py`.

λ	n	$\dim B$	$f^{\lambda^{(\geq 2)}}$	comment
$(3,2,1)$	6	2	2	stable
$(3,2,2,1)$	8	3	3	stable
$(3,3,1,1)$	8	2	2	stable
$(4,3,2)$	9	16	16	stable
$(3,3,2,1)$	9	8	5	non-stable, excess 3
$(4,2,2,1)$	9	12	6	**non-stable,** $\times 2$
$(4,3,1,1)$	9	12	5	non-stable, excess 7
$(3,2,2,2)$	9	4	4	stable
$(3,3,3,1)$	10	5	5	stable
$(3,3,2,2)$	10	6	9	$\dim B < f^{\lambda^{(\geq 2)}}$
$(4,2,2,2)$	10	10	10	stable
$(4,4,1,1)$	10	10	5	**non-stable,** $\times 2$
$(4,3,2,1)$	10	24	16	non-stable, excess 8
$(4,4,2,1)$	11	50	21	non-stable

2.2 The dimension drop at $(3,3,2,2)$

Theorem 1 ($\dim B < f^{\lambda^{(\geq 2)}}$ can occur)**.** *For $\lambda = (3,3,2,2)$ at $n = 10$,*

$$\dim B \;=\; 6, \qquad f^{\lambda^{(\geq 2)}} \;=\; f^{(2,2,1,1)} \;=\; 9.$$

Hence the SYT count is not *an upper bound on $\dim B$ in general.*

Proof (computational). With $\ell = 4$, we form the matrix product $R'_5 R'_6 R'_7 R'_8 R'_9 R'_{10}$ in the Hoefsmit basis of $V_{(3,3,2,2)}$ ($\dim V_\lambda = 768$). Mod-P row reduction of the image gives rank 6. The hook-length formula for $(2,2,1,1)$ gives $f^{(2,2,1,1)} = 9$. □ □

Remark 2 (Qualitative novelty)**.** This is qualitatively different from every previously observed non-stable case. In all prior non-stable cases (e.g. $(4,3,2,1)$ at $n = 10$ with $\dim B = 24 > 16 = f^{(3,2,1)}$), the iterated-image dimension *exceeded* the SYT count — as if the SRD recursion had "opened up" to a larger family of saturated chains. At $(3,3,2,2)$ the chain family is, on the contrary, *linearly dependent*: the SYT count over-counts $\dim B$. We have no structural explanation yet.

2.3 The factor-of-2 sub-family

Observation 3 (Paired equal non-trivial rows)**.** The two cases with $\dim B = 2f^{\lambda^{(\geq 2)}}$ in the scan are

$$(4,2,2,1) \text{ at } n=9: \quad \dim B = 12 = 2\cdot f^{(3,1,1)} = 2\cdot 6,$$
$$(4,4,1,1) \text{ at } n=10: \quad \dim B = 10 = 2\cdot f^{(3,3)} = 2\cdot 5.$$

Both shapes have a pair of equal non-trivial rows (rows $2,3$ of length 2 in the first case; rows $1,2$ of length 4 in the second). All other non-stable cases in the scan have either no such pair, or have their pair embedded in a more elaborate structure, and exhibit no clean integer multiplicity.

Conjecture 4 (Factor-of-2 from row repeats)**.** *Let λ have exactly one pair of equal rows of length $a \geq 2$, with all other rows distinct in length and either of length a (forming the pair) or of length 1. Then $\dim B$ has a factor of 2 relative to $f^{\lambda^{(\geq 2)}}$; more generally, if rows ≥ 2 contain a repeated part with multiplicity k, $\dim B$ may have a hidden multiplicity factor depending on k.*

We do not have computational evidence beyond the two cases above, and the conjecture is offered only as a structural hypothesis to falsify or refine. The earlier non-stable case $(4,3,2,1)$ has all rows of distinct length and shows no integer multiplicity (excess 8, not a clean factor).

2.4 Comparison with the May-13 stability threshold

The May-13 follow-up note (`2026-05-13-decoration-iso-failed.tex`, §5) already established that the stability threshold for the dim formula is shape-dependent and that the matching set is not an up-set in trailing-1-rows. The present finding goes further: the deviation *from* the SYT count is not even sign-controlled. The natural guess $\dim B \geq f^{\lambda^{(\geq 2)}}$ (with equality in the stable regime) is false at $(3,3,2,2)$.

3 Finding 2: B is not JM-semisimple

The Hoefsmit basis diagonalises the Jucys–Murphy elements $J_1,\ldots,J_n$; their joint eigenspaces are exactly the lines $\mathbb{C}v_T$ indexed by SYTs T (this is the content of the seminormal-form theorem for $H_q(S_n)$ at generic q). Hence "B is a direct sum of J-weight spaces" is equivalent to "B is spanned by a subset of the v_T." We disprove this in two stable test cases.

3.1 Methodology

For each test case we compute, in the seminormal basis:

(1) A column-basis matrix $W \in \mathbb{F}_P^{\dim V_\lambda \times \dim B}$ for B, obtained by mod-P row reduction of the product $R'_{\ell+1}\cdots R'_n$.

(2) For each candidate $X \in \{J_1,\ldots,J_n\} \cup \{T_1,\ldots,T_{n-1}\}$, the matrix XW, and the rank of $[W \mid XW]$. If this rank equals $\dim B$, then $XB \subseteq B$; otherwise X fails to preserve B.

(3) For each X that preserves B, the trace of X on B (computed as $\operatorname{tr}(W^+XW)$ where W^+ is a left inverse of W, equivalently by reading off the 2×2 action matrix from the reduced form).

The mod-P rank coincides with the rank over $\mathbb{Q}(q)$ at the specialisation $q=11$ for any non-special q; at $|\lambda| \leq 11$ no special-q obstruction arises. The trace values are likewise polynomials in q that agree with the $\mathbb{Q}(q)$ trace at the specialisation.

3.2 $\lambda = (3, 2, 1, 1, 1)$ at $n = 8$

Here $\ell = 5$, $\dim V_\lambda = 64$, $\dim B = 2$, $f^{\lambda^{(\geq 2)}} = f^{(2,1)} = 2$ — the stable regime.

Theorem 5 (*B* has broad Hoefsmit support)**.** *Let $W \in \mathbb{F}_P^{64\times 2}$ be a column basis of B obtained by mod-P row reduction. The set*

$$\operatorname{supp}(W) \;:=\; \{T \in \mathrm{SYT}(\lambda) : (W_{T,1}, W_{T,2}) \neq (0, 0)\}$$

has cardinality 26*, far exceeding* $\dim B = 2$*. In particular B is* not *the span of any pair of v_T's.*

Proof (computational). Direct count in `jm_test.py` of nonzero rows of the reduced W. □ □

Theorem 6 (Few stabilising JM and Hecke operators)**.** *For $\lambda = (3, 2, 1, 1, 1)$:*

(i) Among $J_1, \ldots, J_8$, only J_1 and J_2 preserve B.

(ii) Among $T_1, \ldots, T_7$, only T_1 and $T_\ell = T_5$ preserve B (cf. the May-13 refutation note).

(iii) T_5 acts on B as $q \cdot I_2$.

Proof (computational). Items (i) and (ii) by the rank test $\mathrm{rk}[W \mid XW] \overset{?}{=} \mathrm{rk}\, W$ for each $X \in \{J_1, \ldots, J_8\} \cup \{T_1, \ldots, T_7\}$; see `check_preserved.py`.

For (iii): on the 2-dim space B, the trace of T_5 was computed as the polynomial $2q$, and the eigenvalues of any T_j on V_λ lie in $\{q, -1\}$ (the quadratic relation $(T_j - q)(T_j + 1) = 0$). The only multiset of two elements from $\{q, -1\}$ with sum $2q$ is $\{q, q\}$, so T_5 has both eigenvalues equal to q on B. Combined with the fact that T_5 is semisimple on V_λ (Hoefsmit), it acts as the scalar q on B. □ □

Remark 7 (Comparison to the May-13 follow-up)**.** The May-13 follow-up note already showed by independent computation that T_5 acts as scalar q and T_1 as scalar -1 on B at $\lambda = (3, 2, 1, 1, 1)$. The new content of Theorem 6 is the JM count: of the eight Jucys–Murphy elements, only the first two preserve B. This is the obstruction to JM-semisimplicity.

3.3 $\lambda = (3, 2, 1, 1)$ at $n = 7$

Here $\ell = 4$, $\dim V_\lambda = 35$, $\dim B = 2$.

Theorem 8 (Same pattern at $n = 7$)**.** *For $\lambda = (3, 2, 1, 1)$:*

(i) The Hoefsmit support of B has cardinality 23 *(out of* 35*).*

(ii) Among $J_1, \ldots, J_7$, only J_1 preserves B.

(iii) Among $T_1, \ldots, T_6$, only $T_\ell = T_4$ preserves B, and it acts as $q \cdot I_2$ (verified by the same trace argument).

Proof (computational). Identical methodology to the $n = 8$ case; `check_preserved.py` and `jm_test.py`. □ □

3.4 Consequences for the iso programme

Corollary 9 (No SYT-subset iso)**.** *If an isomorphism $B \cong V_{\lambda(\geq 2)}$ exists in either of the two test cases above (as modules over some embedded $H_q(S_{n-\ell})$), it cannot be realised by identifying B with the span of a subset of Hoefsmit basis vectors $\{v_T\}$.*

Proof. A subset-spanned subspace is, by Hoefsmit diagonality, simultaneously fixed by all J_k. But for $\lambda = (3,2,1,1,1)$ only J_1, J_2 preserve B and for $\lambda = (3,2,1,1)$ only J_1 does. Both fail to be *all* of $J_1, \dots, J_n$. □ □

This is a substantive obstacle to any combinatorial iso construction ("map chain to SYT, then take the seminormal vector"). The iso, if it exists, requires a basis change that is genuinely q-rational — it mixes Hoefsmit basis vectors at coefficients depending on q — and cannot be read off the SYT bijection.

4 Implications and open questions

1. **The stable picture is delicate.** The May-13 closed-form theorem $\dim B = f^{\lambda(\geq 2)}$ holds robustly in the strict stable regime, but the boundary is intricate. The $(3,3,2,2)$ case shows $\dim B$ can drop *below* the SYT count, meaning the saturated chain family from the SRD recursion becomes linearly dependent. This rules out any clean "$\dim B \leq f^{\lambda(\geq 2)}$ always" statement and any monotone-in-shape inequality.

2. **The factor-of-2 sub-family.** Observation 3 identifies a structural feature (paired equal non-trivial rows) that correlates with a clean integer-multiplicity excess. Whether this generalises to k-fold row repeats $\Rightarrow$ factor of k (or $k!$, or some other combinatorial constant) is open and conjecturally recorded as Conjecture 4.

3. **Induced-module route.** The iso $B \cong V_{\lambda(\geq 2)}$, if it exists in the stable regime, must come from a route other than the naive parabolic embeddings already ruled out and the seminormal-subset identification ruled out above. The most natural remaining candidate is the canonical (up to scalar) $H_q(S_n)$-morphism

$$\psi : \mathrm{Ind}_{H_q(S_\ell) \times H_q(S_{n-\ell})}^{H_q(S_n)} \big(V_{(1^\ell)} \boxtimes V_{\lambda(\geq 2)}\big) \longrightarrow V_\lambda,$$

which exists with multiplicity 1 by the Littlewood–Richardson identity $c^{\lambda}_{(1^\ell),\,\lambda(\geq 2)} = 1$ (worked out in the May-13 follow-up note, §3.1). The question $\mathrm{im}(\psi) \stackrel{?}{=} B$ is open. If the answer is yes, the iso programme reduces to identifying the image of ψ with the iterated R'-image, both being canonical on different sides.

4. $(3,3,2,2)$ **as a test stone.** Any candidate iso theorem must account for the dimension drop at $(3,3,2,2)$. It is unclear whether $(3,3,2,2)$ is "inside" or "outside" the iso theorem's hypothesis: it may be that the iso theorem requires stability *and* that stability for $(3,3,2,2)$ fails in a yet finer sense than the May-13 threshold table indicated. We do not know which.

Question 10 (Image of the induced morphism)**.** Is $\mathrm{im}(\psi) = B^{(\lambda)}_{\ell+1} V_\lambda$ as subspaces of V_λ, at least in the stable regime?

Question 11 (Drop characterisation)**.** For which λ does $\dim B < f^{\lambda(\geq 2)}$? Is $(3,3,2,2)$ the first member of an identifiable family?

5 Honest status

Neither finding is a positive theorem about B; both are obstructions. Finding 1 closes off a natural inequality conjecture; Finding 2 closes off the most combinatorially obvious route to a rep-theoretic iso. The positive content of this note is in the conjectures and the open questions: the induced-module route remains the leading candidate for an iso theorem, and the factor-of-2 phenomenon points to a hidden multiplicity structure in the non-stable regime that is worth chasing.

A Computational verification

All computations in this note were performed mod $P = 100\,003$ at $q = 11$, using the seminormal-matrix infrastructure (`compute_dim_B.py`) and the following scripts:

- `/home/clio/projects/scratch/2026-05-14-non-stable/run_nonstable.py` — computes the full $\dim B$ vs. $f^{\lambda^{(\geq 2)}}$ table for $n \leq 11$ (Finding 1, §2).
- `/home/clio/projects/scratch/2026-05-14-jm-test/jm_test.py` — Hoefsmit-support count of B and JM eigenvalue probes (Finding 2, §3).
- `/home/clio/projects/scratch/2026-05-14-jm-test/check_preserved.py` — preservation tests $\operatorname{rk}[W \mid XW] \stackrel{?}{=} \operatorname{rk} W$ for $X \in \{J_k\} \cup \{T_j\}$ (Finding 2, §3).

Reproducibility: each script writes its summary to a sibling `summary.json` / `output.log`. The mod-P rank equals the rank over $\mathbb{Q}(q)$ for any non-special q (in particular $q = 11$ is non-special for $|\lambda| \leq 11$).